

Highlights

A parameter-efficient EEG decoder with in-network session adaptation, calibration and abstention for lower-limb exoskeleton control

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- A 24,181-parameter EEG decoder emitting class, calibrated confidence and an abstention decision.
- Label-free in-network alignment reduces session drift by 52–84% in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- The adaptation rate is two-sided: two cohorts' optima are opposite, costing 0.06–0.09.
- The admissible rate band is computable from the recording protocol before fitting.

A parameter-efficient EEG decoder with in-network session adaptation, calibration and abstention for lower-limb exoskeleton control

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ARTICLE INFO

Keywords:

brain–computer interface
electroencephalography
test-time adaptation
covariate shift
selective prediction
calibration
lower-limb exoskeleton

ABSTRACT

Background. A decoder that drives a powered lower-limb exoskeleton is fitted once and must keep working for weeks, during which electrode impedance, montage and skin contact change the second-order statistics that every spatial filter depends on. Correcting this shift inside the network, by re-estimating the input covariance at inference and whitening by it without labels, is an attractive design because it needs no target-domain calibration session.

Problem. We show that such a layer carries a failure mode its offline counterpart cannot have. A running estimate tracks whatever is currently streaming. Under a protocol that presents classes in long contiguous blocks, the estimate converges to the currently active class and the layer whitens away the signal it was inserted to protect.

Solution. We present a decoder combining a label-free adaptive alignment layer, a multi-scale power stem that preserves log-power ratios, and a selective head trained under a coverage constraint, and we derive an admissible band for the adaptation rate from two timescales that are measurable from the recording protocol before any model is fitted.

Results. On a 7-subject, 9-session exoskeleton cohort recorded across weeks, the 24,181-parameter configuration reaches 0.901 balanced accuracy with expected calibration error 0.039 and 0.876 accuracy at 90% coverage, at 53% of the parameters of the strongest baseline evaluated in the same pipeline (0.913). The alignment layer reduces session-to-session covariance distance by $52.0 \pm 4.5\%$ and $83.9 \pm 3.9\%$ on two independent cohorts, in 15 of 15 subjects ($p \leq 10^{-4}$), against a no-op control that shows no reduction. On the second cohort the architecture does not transfer: the bare stem scores 0.737 against 0.626 for the full model. Slowing adaptation past the class-block length recovers $+0.090$ to $+0.098$ on all three arms that use alignment and exactly 0.000 on both arms that do not. Applying the same change to the first cohort costs -0.060 , so the two optima are opposite and no single rate serves both protocols.

Significance. The admissible rate band is bounded below by the class-block timescale and above by the drift timescale. Where class blocks outlast drift the band is empty and the layer should be omitted rather than tuned. Because both bounds are properties of the experimental design, the condition is a design-time check rather than a post-hoc diagnosis, and it predicts where in-network adaptation will fail before a model is trained.

1. Introduction

Spinal cord injury leaves roughly 250,000 to 500,000 people per year with impaired or absent lower-limb function, and powered exoskeletons driven by non-invasive brain signals are one route to restoring overground mobility (He, Eguren, Azorín, Grossman, Luu and Contreras-Vidal, 2018; López-Larraz, Trincado-Alonso, Rajasekaran, Peres-Nombela, del Ama, Aranda, Minguez, Gil-Agudo and Montesano, 2016; Do, Wang, King, Chun and Nenadic, 2013). Decoding the wearer’s intent to walk or to stop from scalp EEG is the control problem at the centre of such systems (Contreras-Vidal and Grossman, 2013; Presacco, Goodman, Forrester and Contreras-Vidal, 2011). It is not well described by per-epoch classification accuracy alone, for three reasons that shape the design presented here.

Longitudinal drift The decoder is fitted once and must keep working weeks later. In the primary cohort used here this is literal: the model is fitted on sessions 1–3 and evaluated on sessions 4–9, recorded across weeks.

Electrode montage, impedance and skin contact all change between sessions, and the resulting shift in second-order statistics is the dominant source of non-stationarity in EEG decoding (Shenoy, Krauledat, Blankertz, Rao and Müller, 2006; Krauledat, Tangermann, Blankertz and Müller, 2008; Blankertz, Tomioka, Lemm, Kawanabe and Müller, 2008). Session drift degrades fixed spatial filters and is the reason most BCI systems require a per-session calibration block (Vidaurre, Sannelli, Müller and Blankertz, 2011).

Asymmetric cost A false Walk command moves a person’s legs. Balanced accuracy scores that identically to a missed Walk and cannot distinguish one sustained false activation from fifty scattered false windows. The quantity a wearer experiences is spurious activations per minute of standing, which we report alongside accuracy throughout.

No option to decline A classifier trained only for accuracy must emit a decision for every window. A device that acts on a confidence value needs that value to be calibrated (Guo, Pleiss, Sun and Weinberger, 2017), and needs a principled way to do nothing when the evidence is weak (Chow, 1970; El-Yaniv and Wiener, 2010).

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Existing responses to drift are mostly applied offline in preprocessing. Euclidean Alignment whitens each recording by the inverse square root of its mean covariance (He and Wu, 2020); Riemannian Procrustes analysis and parallel transport generalise this to the manifold of symmetric positive-definite (SPD) matrices (Rodrigues, Jutten and Congedo, 2019; Yair, Ben-Chen and Talmon, 2019; Zanini, Congedo, Jutten, Said and Berthoumieu, 2018). All of these estimate the alignment statistic over a complete recording, which means the target recording must exist before the decoder can run on it. Test-time adaptation removes that requirement in computer vision by updating normalisation statistics or parameters online (Wang, Shelhamer, Liu, Olshausen and Darrell, 2021; Schneider, Rusak, Eck, Bringmann, Brendel and Bethge, 2020; Nado, Padhy, Sculley, D'Amour, Lakshminarayanan and Snoek, 2020), and recent work shows that such methods degrade when the test stream is temporally correlated rather than i.i.d. (Gong, Jeong, Kim, Kim, Shin and Lee, 2022; Niu, Wu, Zhang, Wen, Chen, Zhao and Tan, 2023; Wang, Fink, Van Gool and Dai, 2022). EEG data acquired under a blocked experimental protocol is temporally correlated by construction, which is the connection this paper makes precise.

Our contributions are:

1. We introduce an adaptive alignment layer that estimates and applies a whitening transform without labels and continues to update at inference, and we verify that its output reaches the classifier rather than being undone by the following normalisation (Section 4.1).
2. We identify a failure mode specific to in-network alignment: the running estimate tracks the currently streaming class when class blocks exceed the estimator's memory, and the layer then removes the discriminative signal (Section 6.7).
3. We derive an admissible band for the adaptation rate, bounded below by the class-block timescale and above by the drift timescale, and show the condition is two-sided by testing it in the direction where we predicted no effect (Proposition 1, Section 6.8).
4. We validate the mechanism separately from the task, reporting a $52.0 \pm 4.5\%$ and $83.9 \pm 3.9\%$ reduction in session-to-session covariance distance in 15 of 15 subjects against a no-op control (Section 6.1).
5. We report an external validation that fails, with the ordering of arms inverting between cohorts, and trace the inversion to the rate condition rather than to sample size or implementation (Section 6.4).

We make no accuracy claim over published decoders. In the pipeline where both were evaluated, ATCNet (Altaheri, Muhammad and Alsulaiman, 2023) scores 0.913 against 0.901 for our configuration, and we report that as a loss in Section 6.3.

2. Related work

2.1. Compact convolutional decoders for EEG

ShallowFBCSPNet and Deep4Net established that a square $\rightarrow$ pool $\rightarrow$ log pipeline over learned spatial filters recovers the band-power features that filter-bank common spatial patterns computes explicitly (Schirrmester, Springenberg, Fiederer, Glasstetter, Eggensperger, Tangermann, Hutter, Burgard and Ball, 2017). EEGNet reduced the parameter count by an order of magnitude using depthwise-separable convolutions (Lawhern, Solon, Waytowich, Gordon, Hung and Lance, 2018), and EEGNeX refined the same family (Chen, Teng, Chen, Pan and Geyer, 2024). These architectures work well within a session and are the standard baselines in the field (Roy, Banville, Albuquerque, Gramfort, Falk and Faubert, 2019; Craik, He and Contreras-Vidal, 2019). Their limitation for our setting is that the learned spatial filters are fixed at training time, so a montage or impedance change between sessions is not compensated at all. We respond by placing an adaptive whitening stage before the spatial filters, so that what the filters see is referred to a covariance estimated on the current recording (Section 4.1).

2.2. Attention-based decoders

ATCNet combines convolutional feature extraction, sliding-window multi-head attention and temporal convolution, and is the strongest baseline in our comparison (Altaheri et al., 2023). EEG Conformer pairs a ShallowFBCSP-style stem with a transformer encoder (Song, Zheng, Liu and Gao, 2023), and EEG-TCNet applies temporal convolutions to the same problem (Ingolfsson, Hersche, Wang, Kobayashi, Cavigelli and Benini, 2020). We note explicitly that the stem-plus-transformer combination is already published: an independent implementation of it in our pipeline reproduced EEG Conformer's score rather than improving on it (0.848 against 0.843), and we therefore claim no novelty for that pairing. What these models share with the convolutional family is a fixed front end, which is the property the present work targets.

2.3. Alignment and transfer in BCI

Riemannian methods treat the spatial covariance as a point on the SPD manifold and classify with geodesic distances (Barachant, Bonnet, Congedo and Jutten, 2012, 2013; Congedo, Barachant and Bhatia, 2017). For transfer, Euclidean Alignment whitens each subject or session by $\mathbf{M}^{-1/2}$ where $\mathbf{M}$ is its mean covariance (He and Wu, 2020); Riemannian Procrustes analysis adds rotation and scaling (Rodrigues et al., 2019), and parallel transport moves covariances between tangent spaces (Yair et al., 2019; Zanini et al., 2018). These methods are effective and label-free, and reviews report them as the strongest general-purpose transfer tools for EEG (Lotte, Bougrain, Cichocki, Clerc, Congedo, Rakotomamonjy and Yger, 2018; Wu, Xu and Lu, 2022). Their limitation is operational rather than statistical: the alignment statistic is computed over a whole recording, so the recording must be complete before decoding starts, and

within-recording drift is not tracked. We respond by moving the estimator inside the network as a layer that updates online, and we then show what that move costs (Section 6.7).

2.4. Test-time adaptation

Updating normalisation statistics on the test distribution improves robustness to corruption in vision (Schneider et al., 2020; Nado et al., 2020), and Tent extends this to updating affine parameters by entropy minimisation (Wang et al., 2021). Test-time training adds a self-supervised objective at inference (Sun, Wang, Liu, Miller, Efros and Hardt, 2020). A recurring finding in this literature is that these methods assume the test stream is i.i.d., and degrade when it is not: NOTE addresses temporally correlated streams by maintaining a class-balanced reservoir (Gong et al., 2022), SAR stabilises adaptation under distribution shift with small or imbalanced batches (Niu et al., 2023), and CoTTA and LAME address error accumulation in continual adaptation (Wang et al., 2022; Boudiaf, Mueller, Ben Ayed and Bertinetto, 2022). Our contribution relative to this line is quantitative rather than conceptual: we show that in a blocked EEG protocol the relevant comparison is between the estimator's effective memory and the duration of a single-class block, that this predicts the threshold location, and that the resulting constraint is two-sided because the same estimator is also what tracks the drift the layer exists to remove.

2.5. Calibration and selective prediction

Modern networks are overconfident, and expected calibration error (Naeini, Cooper and Hauskrecht, 2015; Guo et al., 2017) degrades further under distribution shift (Ovadia, Fertig, Ren, Nado, Sculley, Nowozin, Dillon, Lakshminarayanan and Snoek, 2019), which is exactly the regime a longitudinally deployed decoder operates in. Selective classification allows a model to abstain (Chow, 1970; El-Yaniv and Wiener, 2010), and SelectiveNet trains the classifier and the selection function jointly under a coverage constraint (Geifman and El-Yaniv, 2019, 2017). We use SelectiveNet as published and claim no contribution to it; we include it because a decoder that must not move a wearer's legs on weak evidence needs the reject option, and because it is the only component in our ablation whose benefit transfers across both cohorts.

2.6. EEG correlates of gait and exoskeleton control

Walking modulates cortical activity in a gait-phase-locked manner (Gwin, Gramann, Makeig and Ferris, 2011; Seeber, Scherer, Wagner, Solis-Escalante and Müller-Putz, 2014), and treadmill walking can be decoded from scalp EEG (Presacco et al., 2011; Luu, Nakagome, He and Contreras-Vidal, 2017). Robot-assisted gait modulates midline sensorimotor rhythms (Wagner, Solis-Escalante, Grieshofer, Neuper, Müller-Putz and Scherer, 2012). The intent signal our decoder targets is event-related desynchronisation in the mu and beta bands, a power decrease over sensorimotor cortex preceding and accompanying movement (Pfurtscheller and Lopes da Silva, 1999; Pfurtscheller, Brunner, Schlögl and Lopes da Silva, 2006). This motivates a representation in

which log-power ratios are preserved (Section 4.2), because ERD is defined as a relative power change.

3. Problem formulation

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ be a window of C channels and T samples with label $y \in \{\text{Stop}, \text{Walk}\}$. A recording session s supplies windows drawn from $p_s(\mathbf{X}, y)$. We assume the conditional $p(y | \mathbf{X})$ is stable across sessions but the marginal $p_s(\mathbf{X})$ is not, which is the covariate-shift assumption standard in this literature (Sugiyama, Krauledat and Müller, 2007; Shenoy et al., 2006). Concretely we take the shift to act primarily on the spatial covariance $\Sigma_s = \mathbb{E}_{p_s}[\mathbf{X}\mathbf{X}^\top]/T$.

The decoder is fitted on sessions S_{fit} and evaluated on disjoint later sessions S_{test} , with no labels available from S_{test} at any point. This is the constraint that rules out per-session recalibration and motivates a label-free correction.

We quantify session drift by the affine-invariant Riemannian distance between SPD matrices,

$$\delta(\mathbf{A}, \mathbf{B}) = \left\| \log(\mathbf{A}^{-1/2} \mathbf{B} \mathbf{A}^{-1/2}) \right\|_F, \quad (1)$$

rather than the Frobenius distance, because covariances do not form a vector space and the Frobenius distance is dominated by overall scale, which would flatter any whitening operation (Congedo et al., 2017; Barachant et al., 2012).

Two timescales govern the analysis in Section 6.7. Let τ_{blk} be the median duration, in windows, of a contiguous single-class block in the recording protocol, and let τ_{drift} be the timescale over which Σ_s changes appreciably. Both are properties of the experimental design and the recording hardware, not of the model.

4. Method

The decoder maps a window to a class posterior and a selection score. Its four components are shown in Fig. 1 with exact parameter counts.

4.1. Adaptive alignment

The layer maintains a running mean spatial covariance $\mathbf{M}$, applies $\mathbf{M}^{-1/2}$, and interpolates between the aligned and the raw signal through a learned scalar gate $g \in [0, 1]$:

$$\mathbf{X}' = g \mathbf{M}^{-1/2} \mathbf{X} + (1 - g) \mathbf{X}. \quad (2)$$

Each window's covariance is normalised by its own trace before being accumulated, so that the estimate is an average of shapes rather than an average weighted by power:

$$\mathbf{M} \leftarrow (1 - m) \mathbf{M} + \frac{m}{N} \sum_{n=1}^N \frac{\mathbf{C}_n}{\text{tr}(\mathbf{C}_n)/C}, \quad \mathbf{C}_n = \frac{\tilde{\mathbf{X}}_n \tilde{\mathbf{X}}_n^\top}{T}, \quad (3)$$

where $\tilde{\mathbf{X}}_n$ is the mean-centred window, N the batch size and m the momentum. Without the trace normalisation the estimate is power-weighted. On a cohort whose majority class carries 1.52× the power of the minority class, the unnormalised estimate sits at Riemannian distance 0.306 from

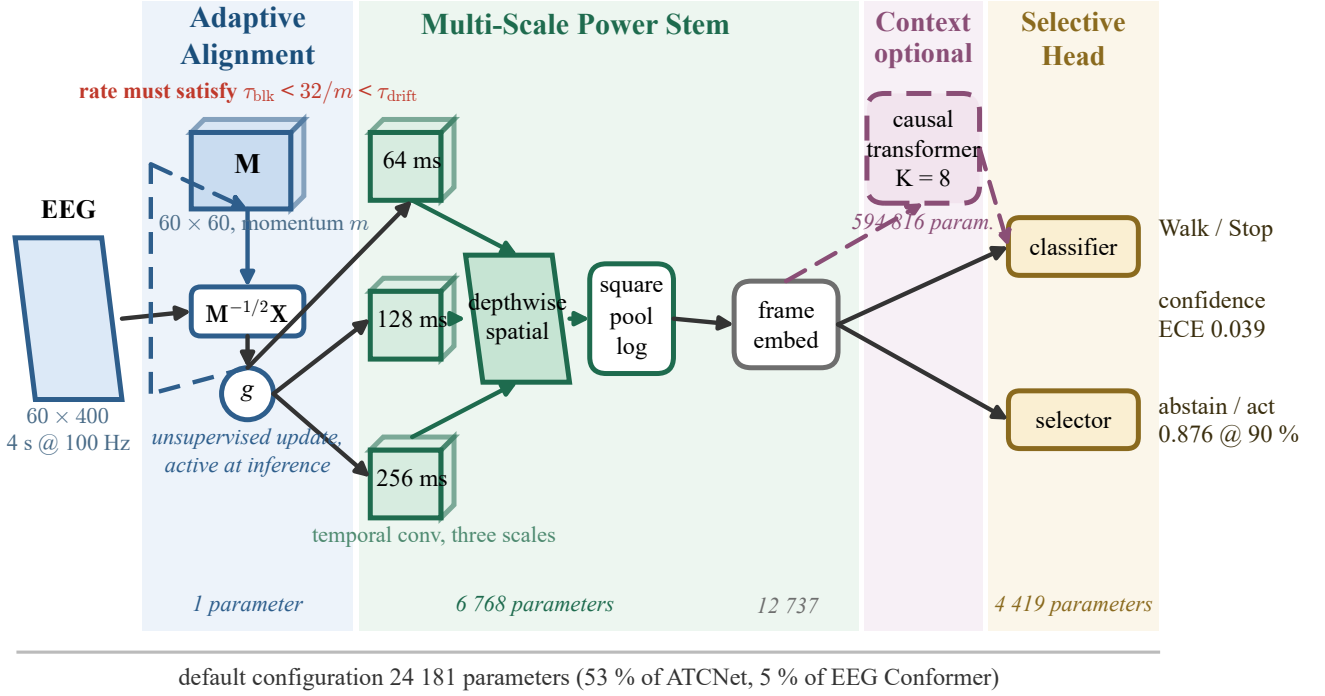


Figure 1: The decoder. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) uses no labels and remains active at inference, which is what separates it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594,816 of the 618,997 parameters of the full configuration, so the default configuration omits it and uses 24,181. Parameter counts are exact and sum to the total.

the majority-class covariance, that is, it is the majority class; the normalised estimate sits at 3.899 and 3.194 from the two class covariances, between them. Section 6.6 reports what correcting this did and did not achieve.

Two properties separate this layer from batch normalisation (Ioffe and Szegedy, 2015). The update in (3) uses only the covariance of incoming windows and therefore runs on unlabelled test data. And the estimate continues to update in evaluation mode, whereas BatchNorm freezes its running statistics at test time. Freezing is correct when train and test share a distribution; here they are weeks apart, and freezing discards the adaptation the layer exists to provide.

The layer’s output reaches the classifier A whitening stage placed before a normalisation layer invites the objection that the normalisation simply undoes it. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36 % in relative magnitude and their correlation structure differs by 0.108, so the transform is not absorbed downstream.

4.2. Multi-scale power stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, are followed by square → pool → log, after Schirrneister et al. (2017).

Amplitude normalisation precedes the squaring so that log-power ratios survive: scaling a window by s shifts its log-power by $2 \log s$, a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio in which ERD is defined (Pfurtscheller and Lopes da Silva, 1999).

This ordering is load-bearing rather than cosmetic, and the evidence is a dose–response across representations that differ in how much marginal power they carry (Table 1). Removing per-window amplitude normalisation is worth +0.134 and +0.174 on band-power features, which are pure marginal power; −0.016 and +0.029 on tangent-space features, which carry power only through the covariance diagonal; and −0.004 and +0.000 on a correlation-only representation, which is scale-free by construction and therefore cannot respond. A representation that carries no marginal power shows no effect, which is what identifies the mechanism rather than merely demonstrating it.

A separate sweep over where normalisation is applied in the network confirms the ordering at the architecture level: on cohort A with 4 s windows, per-window normalisation scores 0.849, global per-recording normalisation 0.878, and no normalisation 0.864, so the global setting used here is the best of the three and per-window normalisation is the worst.

Table 1

Amplitude-normalisation dose–response. z_1 applies per-window z -scoring before feature extraction, z_0 omits it. The effect scales with how much marginal power the representation carries, and vanishes for the scale-free representation.

Representation	Power content	z_1	z_0	Δ
<i>Cohort A</i>				
Band power	full	0.625	0.760	+0.134
Tangent space	partial	0.762	0.746	−0.016
Correlation	none	0.738	0.734	−0.004
<i>Cohort B</i>				
Band power	full	0.574	0.748	+0.174
Tangent space	partial	0.675	0.704	+0.029
Correlation	none	0.701	0.702	+0.000

4.3. Causal cross-epoch context

A causal transformer encoder (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017) over a sequence of $K = 8$ epochs at stride 3 (a 14.5 s span), following the epoch-sequence formulation standard in sleep staging (Phan, Andreotti, Cooray, Chén and De Vos, 2019; Phan, Chén, Tran, Koch, Mertins and De Vos, 2021; Supratak, Dong, Wu and Guo, 2017). All masks are causal, so only present and past information is used, which is the only variant a worn device can run. Measured state persistence in cohort A supports a sequence model: the self-transition probability is 0.984 for Stop and 0.955 for Walk at a 0.5 s step, corresponding to dwell times of 12 to 32 s.

4.4. Selective head

A classifier paired with a selection function trained under a coverage constraint (Geifman and El-Yaniv, 2019), with a full-coverage auxiliary term so that the backbone continues to learn from every sample rather than only from the covered region. We use the formulation as published.

5. Experimental setup

5.1. Data

Cohort A (exoskeleton, OpenNeuro ds007788). Seven participants, nine sessions each, recorded across weeks while walking in a powered lower-limb exoskeleton (Contreras-Vidal and Grossman, 2013). Sixty-channel active EEG at 100 Hz; walk/stop labels are taken from the exoskeleton controller state. Windows of 4 s at 0.5 s stride give approximately 5031 fitting windows per participant. The model is fitted on sessions 1–3 and evaluated on sessions 4–9.

Cohort B (treadmill MoBI). Eight participants, three trials each, 64-channel EEG at 100 Hz, walk/stand from treadmill phase (Luu et al., 2017). Fitted on trial 1 and evaluated on trials 2–3, with 2 s windows. The cohort is from an independent laboratory with different sensors, a different task framing, and reversed class balance: Stop is the minority class at approximately 12 % here and the majority class in cohort A.

Table 2

Contiguous single-class block structure. τ_{blk} is the median block duration in windows.

Cohort	Blocks	τ_{blk}	Longest
A (exoskeleton)	109	18	126
B (treadmill)	7	309	2211

Table 3

Training configuration, identical across all arms and both cohorts except where the cohort definition differs.

Setting	Value
Band-pass / resample	0.5–40 Hz, 100 Hz
Artefact rejection	none (deliberate)
Amplitude normalisation	global, per recording
Window / stride	4 s / 0.5 s (A), 2 s (B)
Context length K , stride	8 epochs, 3 (14.5 s span)
Optimiser	Adam (Kingma and Ba, 2015), $\text{lr } 10^{-3}$
Batch size N	32
Epochs	30
Loss	cross-entropy, inverse-frequency weighted
Selective coverage target	0.90
Adaptation momentum m	0.2 (swept in §6.7)
Model selection	best held-out balanced accuracy

The two protocols differ sharply in temporal structure, which Section 6.7 shows to be decisive (Table 2).

5.2. Preprocessing and training

Signals are band-pass filtered to 0.5–40 Hz and resampled to 100 Hz using MNE-Python (Gramfort, Luessi, Larson, Engemann, Strohmeier, Brodbeck, Goj, Jas, Brooks, Parkkonen and Hämäläinen, 2013). No artefact rejection, ICA decomposition or channel interpolation is applied, so that the alignment layer is evaluated on the shift it is meant to correct rather than on a cleaned signal. Amplitude normalisation is applied globally per recording, before the squaring stage, for the reason given in Section 4.2.

Table 3 lists the training configuration. The adaptation momentum is $m = 0.2$ unless stated otherwise; Section 6.7 treats m as the object of study. Model selection uses balanced accuracy on a held-out split of the fitting sessions, evaluated at the best epoch by that criterion, which is conventional but upward-biased on small folds (Section 8).

Backbone screening Before settling on the stem described in Section 4.2, we screened 19 published decoders from the *braindecode* library on a three-participant subset of cohort A (sub-01, sub-03, sub-06) under identical preprocessing (Table 4). The spread is wide, from chance to 0.827, and several architectures that perform well on motor-imagery benchmarks sit near chance on this task. We report this as a screening result on three participants rather than a benchmark, and it is not comparable to Table 11, which uses all seven participants.

Table 4

Backbone screening on three participants of cohort A, single seed, identical preprocessing. A screening result, not a benchmark; the full-cohort comparison is Table 11. EEGNeX failed with a numerical error under this configuration and is recorded as such.

Model	Acc	Model	Acc
EEG Conformer	0.827	FBCNet	0.685
TSception	0.789	ShallowFBCSPNet	0.655
MSVTNet	0.778	SincShallowNet	0.627
EEGSimpleConv	0.763	IFNet	0.600
EEGInceptionMI	0.758	EEGNet	0.573
FBLightConvNet	0.751	Deep4Net	0.536
FBMSNet	0.690	CTNet	0.532
SCCNet	0.698	BDTCN	0.521
		EEGITNet	0.509
		EEG-TCNet	0.500
		EEGNeX	error

Table 5

Leakage sensitivity. A known fraction f of the movement signal is mixed into the EEG before decoding; $f = 0$ is the unmodified recording. The observed clean accuracy places an upper bound on real contamination.

f	0	0.001	0.005	0.01	0.05	0.5
Acc	0.763	0.819	0.874	0.892	0.913	0.917

5.3. Outcome measures

Balanced accuracy for comparability with the literature; expected calibration error (ECE) with 10 equal-width bins (Naeini et al., 2015), because a device acting on a confidence value needs that value to mean something; accuracy at 90 % coverage, the trained selective operating point; spurious activations per minute of standing; and detection latency, reported alongside the benefit it buys.

Leakage guard Unconditioned nuisance-decodability is confounded with accuracy. In our pipeline a decoder that uses no movement information whatever measures $R^2 = +0.863$ against the movement signal, because both the decoder output and the nuisance are driven by the class label. Movement leakage is therefore measured conditional on the class label throughout, by centring the nuisance features within class before regression.

We bound the residual contamination by injecting a known fraction f of the movement signal into the EEG and measuring the accuracy response (Table 5). Accuracy rises monotonically with f and the observed clean accuracy corresponds to $f < 0.005$, so any genuine movement contamination accounts for under 0.5 % of the signal power the decoder uses.

5.4. Statistical treatment and measurement noise

Drift reductions are tested with a two-sided paired t -test over participants, pairing each participant's aligned distance against its raw distance. Accuracy comparisons are reported

Table 6

Run-to-run spread on arms where the manipulated variable is inert. Because alignment is disabled, neither momentum nor estimator version can affect these arms, so all variation is run-to-run.

Arm	Transformer	n	Spread	SD
Stem only, cohort B	no	3	0.0000	0.0000
No-align, cohort A	yes	3	0.0069	0.0039
No-align, cohort B	yes	4	0.0107	0.0049

Table 7

Two-seed replication on cohort A, original estimator. Seed changes the fit/test split assignment. The spread is the quantity that governs whether two models can be separated.

Arm	Seed 0	Seed 1	Mean	Spread
align + gate	0.901	0.882	0.892	0.020
align + ctx + gate	0.878	0.866	0.872	0.012

as the mean over participants with the between-subject standard deviation; we do not perform paired subject-level significance tests on the accuracy comparisons, and Section 8 states why this matters.

Two ablation arms disable alignment entirely and therefore never invoke the running estimator. For them the momentum and the estimator version are inert, so every run is a true repeat and they measure the noise floor directly (Table 6).

The stem-only arm reproduces 0.7375 bit-for-bit across three runs; the arms containing the cross-epoch transformer do not, and the attention path is the only structural difference between them. We attribute the variation to non-deterministic reductions in scaled dot-product attention on GPU.

Run-to-run variation at a fixed split seed is a lower bound on the relevant noise. Repeating the two strongest arms at a second split seed (Table 7) gives a between-seed spread of 0.012 to 0.020, about twice the fixed-seed figure, which is the quantity that governs whether two models can be ranked.

Three consequences follow. Any arm containing the transformer carries approximately 0.010 run-to-run noise at fixed seed and up to 0.020 across seeds, so differences below 0.02 are not interpreted anywhere in this paper. The effects we do interpret (+0.090 to +0.098 in Table 15, −0.060 in Table 16) are three to five times the between-seed spread. And an exact null in the deterministic arm is stronger evidence than one merely within noise, because that arm has no noise to hide in.

Pipeline separation Our architecture and the ATCNet variants were evaluated in one training pipeline; the remaining published baselines were evaluated in an earlier one. ATCNet was run in both and scores 0.913 and 0.881 respectively, so the two pipelines differ by 0.032 for an identical model, which is larger than most differences of

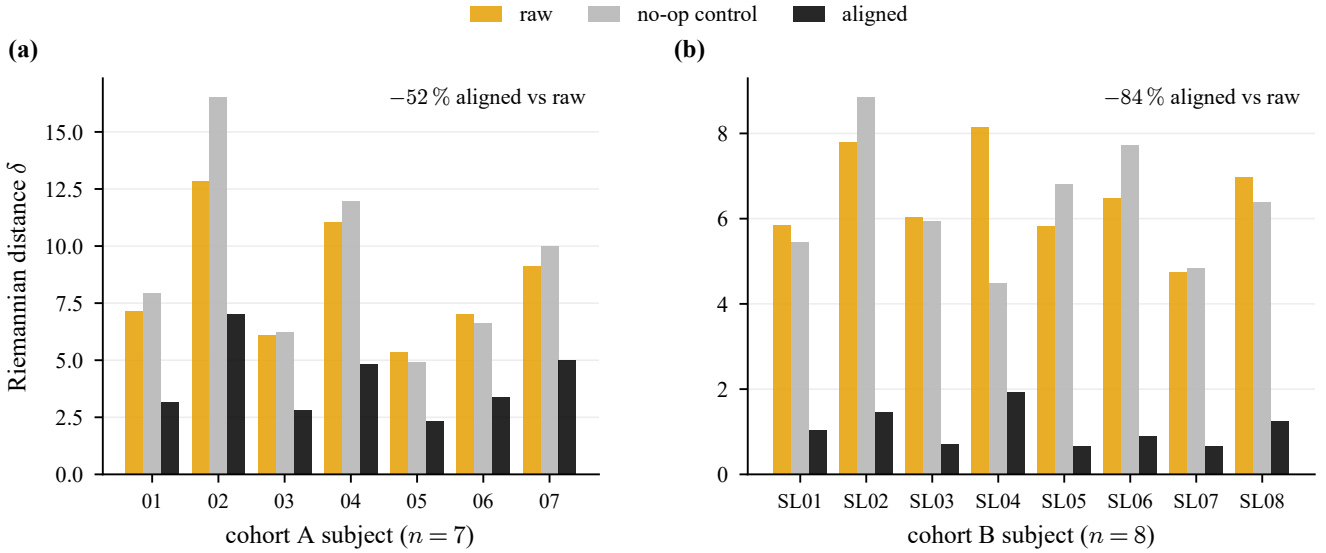


Figure 2: Session-drift reduction per participant. Bars are the Riemannian distance δ between the fitting-session mean covariance and the held-out session mean, for the raw signal, the momentum-0 no-op control, and the aligned signal. (a) cohort A, $n = 7$; (b) cohort B, $n = 8$. Distance falls in every participant of both cohorts; the no-op control, which shares every code path but never updates its estimate, shows no reduction.

Table 8

Session-drift reduction. The no-op control shares every code path but never updates its estimate. Reduction is mean $\pm$ SD over participants; p from a two-sided paired t -test.

Cohort	Raw	Aligned	No-op	Reduction
A ($n=7$)	8.372	4.082	9.173	$52.0 \pm 4.5\%$
B ($n=8$)	6.479	1.074	6.309	$83.9 \pm 3.9\%$

interest in this paper. We therefore report the two groups separately and never compare across them.

6. Results

6.1. The drift mechanism

We measure the Riemannian distance (1) between the fitting-session mean covariance and each held-out session's, before and after the layer (Fig. 2, Table 8).

Drift is reduced in 15 of 15 participants ($t = -9.14$, $p = 1 \times 10^{-4}$ for cohort A; $t = -20.59$, $p < 1 \times 10^{-5}$ for cohort B), with per-participant reductions spanning 45.2 to 56.1 % on cohort A and 76.3 to 88.6 % on cohort B (Table 9). No participant regresses, and the reduction is not carried by one or two outliers. The no-op control shows no reduction in either cohort, which establishes that the effect comes from adaptation rather than from the whitening operation or from the measurement itself. The mechanism works as designed, and Section 6.4 shows that this does not settle whether it helps the task.

6.2. Component ablation, cohort A

The acc@90 value for the *align + ctx* arm is withheld because with the selection head disabled the model emits

Table 9

Per-participant session drift. δ is the Riemannian distance (1) between the fitting-session mean covariance and the mean over held-out sessions. The no-op control shares every code path but never updates its estimate. Every participant improves.

Participant	Raw	No-op	Aligned	Reduction
<i>Cohort A, 6 held-out sessions each</i>				
sub-01	7.154	7.938	3.144	56.1 %
sub-02	12.832	16.522	7.013	45.3 %
sub-03	6.092	6.228	2.827	53.6 %
sub-04	11.035	11.981	4.840	56.1 %
sub-05	5.336	4.906	2.346	56.0 %
sub-06	7.016	6.644	3.399	51.5 %
sub-07	9.139	9.990	5.007	45.2 %
<i>Cohort B, 2 held-out trials each</i>				
SL01	5.845	5.450	1.039	82.2 %
SL02	7.795	8.850	1.451	81.4 %
SL03	6.030	5.934	0.713	88.2 %
SL04	8.158	4.486	1.933	76.3 %
SL05	5.823	6.799	0.666	88.6 %
SL06	6.482	7.722	0.888	86.3 %
SL07	4.738	4.849	0.653	86.2 %
SL08	6.966	6.378	1.250	82.1 %

near-constant confidence, so ranking by it selects the first 90 % of the test set in recording order. That number would be an ordering artefact, not a selective accuracy.

Alignment is necessary, not additive With alignment removed, the context and selection modules together contribute between 0.000 and 0.007: the no-align arm spans 0.827–0.834 across repeats (Table 6) while the stem-only arm is deterministic at 0.827. Against the 0.010 noise floor

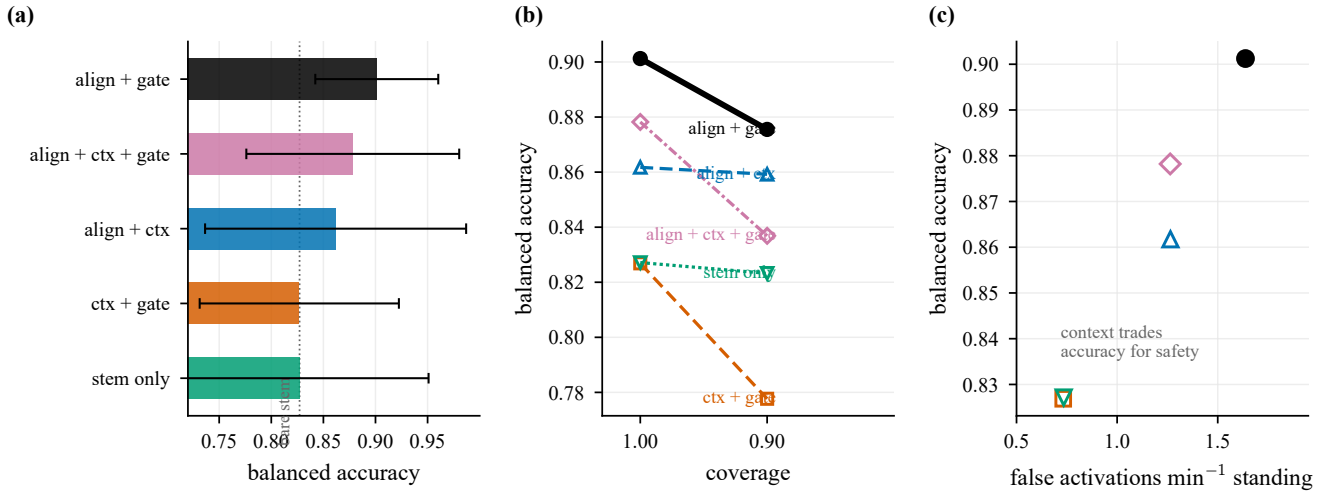


Figure 3: Component ablation on cohort A, seed 0. (a) Balanced accuracy by arm; error bars are ± 1 SD across the 7 participants, not across seeds. (b) Accuracy recovered by declining the least-confident 10 % of windows. (c) Balanced accuracy against spurious activations per minute of standing, showing the cross-epoch context as a selectable operating point.

Table 10

Component ablation, cohort A, single seed. Accuracy is the mean over 7 participants with the between-subject SD. Acc@90 is accuracy at 90 % coverage; FA/min is spurious activations per minute of standing. All cells use the original estimator so that arms are mutually comparable.

Components	Acc $\pm$ SD	ECE	Acc@90	FA/min
align + gate	0.901 $\pm$ 0.059	0.039	0.876	1.64
align + ctx + gate	0.878 $\pm$ 0.102	0.074	0.837	1.26
align + ctx	0.862 $\pm$ 0.125	0.072	—	1.26
ctx + gate	0.827 $\pm$ 0.096	0.091	0.778	0.73
stem only	0.827 $\pm$ 0.124	0.082	0.823	0.73

that is indistinguishable from nothing, and an order of magnitude below alignment's own contribution of +0.028 to +0.051 depending on estimator version. The modules do not stack; they depend on the alignment stage to be useful at all.

Cross-epoch context is an operating point Adding it costs 0.023 accuracy and raises ECE from 0.039 to 0.074, while reducing spurious activations from 1.64 to 1.26 per minute, a 23 % reduction. We therefore report it as a selectable safety mode rather than as part of the default configuration. Removing it also removes 594,816 of the model's 618,997 parameters, which is what makes the default configuration compact.

6.3. Comparison with published decoders

In the pipeline where both were measured, ATCNet scores 0.913 against 0.901 at seed 0 and 0.892 for our two-seed mean (Table 7). The gap of 0.012 to 0.021 sits inside the 0.020 between-seed spread of Section 5.4, so two seeds cannot separate the two models. We report it as a loss rather than as a tie because the point estimate favours ATCNet in both seeds we ran.

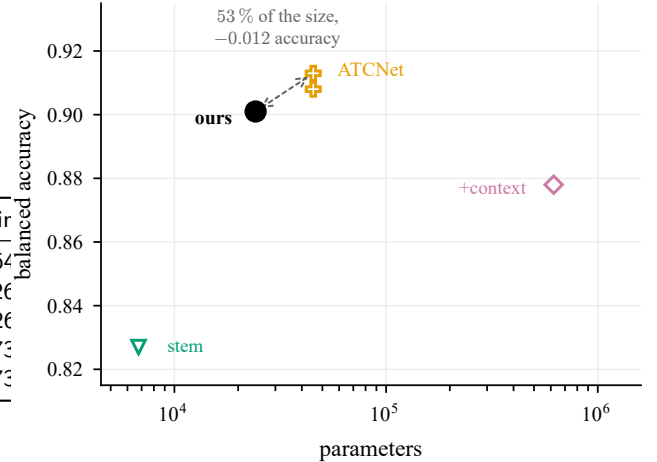


Figure 4: Balanced accuracy against parameter count on cohort A, single seed, pipeline C only. The proposed configuration sits 0.012 below ATCNet at 53 % of its parameters. No point from the other pipeline appears here.

The modules do not transfer to another backbone If the selection and context modules were generally useful, attaching them to a stronger backbone should help. They do not: stock ATCNet with our cross-epoch context and selective head scores 0.894 against 0.913 for stock ATCNet alone, a loss of 0.018 (Table 12). The modules improve calibration (ECE 0.058, and an abstention point at 0.849) and cost accuracy, which is the same trade seen within our own architecture in Table 10, and it is a trade, not an improvement.

What the configuration offers instead is parity with the strongest baseline at 53 % of its parameters and 5 % of EEG Conformer's (Fig. 4), together with calibrated confidence (ECE = 0.039) and a trained abstention operating point (0.876 at 90 % coverage), neither of which the baselines provide. We note that the baselines were not compressed

Table 11

Cohort A comparison. Groups are separated by training pipeline and must not be compared across the rule; ATCNet appears in both and differs by 0.032. Pipeline C is single seed; pipeline F is the mean of 2 seeds with the per-seed range in brackets.

Model	Params	Bal.
<i>Pipeline C (this work), seed 0</i>		
ATCNet (Altaheri et al., 2023)	45,280	0.91
ATCNet, rate-matched kernels	45,280	0.90
Ours, align + gate	24,181	0.90
Ours, align + ctx + gate	618,997	0.87
Ours, stem only	6,768	0.82
<i>Pipeline F (earlier), mean of 2 seeds</i>		
ATCNet (Altaheri et al., 2023)	45,280	0.87
EEGNeX (Chen et al., 2024)	58,082	0.87
ShallowFBCSPNet (Schirrmester et al., 2017)	97,120	0.867 [0.856–0.877]
EEG Conformer (Song et al., 2023)	440,706	0.843 [0.834–0.852]

Table 12

Attaching our two modules to a stronger backbone, cohort A, single seed. Stock ATCNet has no selection head, so no abstention operating point is defined for it.

Model	Acc	ECE	Acc@0.90
Stock ATCNet (Altaheri et al., 2023)	0.913	—	—
ATCNet + ctx + selective head	0.894	0.058	0.849
ATCNet, rate-matched kernels	0.908	—	—

Table 13

Both cohorts, single seed, original estimator.

Components	Cohort A	Cohort B
align + gate	0.901	0.581
align + ctx + gate	0.878	0.626
align + ctx	0.862	0.606
ctx + gate	0.827	0.792
stem only	0.827	0.737

for this comparison, so the parameter axis reports what the published architectures cost, not their achievable minimum.

6.4. External validation: the ordering inverts

On cohort B the bare stem outperforms the full model, 0.737 against 0.626, and the best cohort-A configuration is the worst cohort-B one. The ordering does not merely compress; it inverts, and the three arms that use alignment fall below both arms that do not. We report this in full rather than restricting the paper to the cohort on which the architecture succeeds, and the remainder of the results explains it, because the explanation turns out to be the more transferable contribution.

6.5. Interventions tested and rejected

Six architectural interventions were tested before the configuration reported here and rejected by their own controls (Table 14). We report them because each control is what

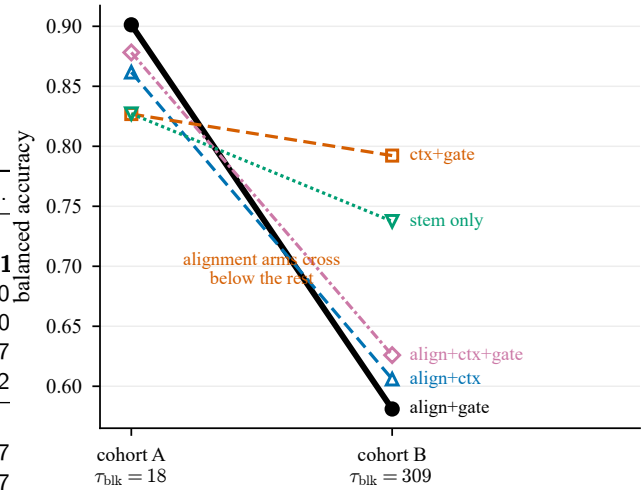


Figure 5. The same five arms on both cohorts, single seed. Solid heavy black is the proposed configuration; the other alignment-using arms are dashed in colour, and the two alignment-free arms are dashed and dotted. The three alignment arms cross below both alignment-free arms, so the ordering inverts rather than compressing.

Table 14

Interventions tested and rejected, each by a pre-specified control. Cohort A unless stated. The motion-cancellation row is the clearest case: the module scored 0.905 with the inertial channel present and chance with it zeroed, so it was an inertial classifier, not an EEG one.

Intervention	Control	Outcome
Motion cancellation	inertial input zeroed	0.905 → 0.501
	inertial input shuffled	0.905 → 0.529
Amplitude side-channel	label-shuffled control	control matches
ShallowFBCSP × ATCNet	vs EEG Conformer	0.848 vs 0.843
Capacity scaling	19k–441k parameters	no trend
MRCP dual-band pathway	vs ERD alone	0.742 vs 0.775
Within-epoch attention	ablate the attention	+0.024 to remove

distinguishes a mechanism from a correlation, and because two of them looked convincing until the control was run.

The movement-related cortical potential (MRCP) pathway deserves a sentence because it is a physiologically motivated design that failed cleanly. Decoding from the 0.1–3 Hz onset-locked band alone gives 0.604; from the mu and beta ERD band alone, 0.775; combining the two, 0.742. The paired difference between the combination and the better single band is -0.034 , so on single-trial data the low-frequency pathway adds noise rather than information, and the architecture uses the ERD band only.

6.6. A rejected explanation for the inversion

Our first diagnosis was the power-weighted estimator described in Section 4.1. The defect is real and measurable: on cohort B, where the majority class carries $1.52\times$ the power, the un-normalised running estimate sat at Riemannian distance 0.306 from the walk covariance, which is to say it had become the majority class. The per-window trace

Table 15

Momentum sweep, cohort B, all five arms, single seed. “At crossing” is $m=0.20 \rightarrow 0.05$, where the estimator memory passes the 309-window class block. Every arm that uses alignment gains; neither arm that omits it responds.

Arm	0.20	0.05	0.01	At crossing	Aligns
align+ctx+gate	0.626	0.724	0.723	+0.098	yes
align+gate	0.581	0.671	0.696	+0.090	yes
align+ctx	0.606	0.700	0.726	+0.094	yes
ctx+gate	0.792	0.782	0.783	−0.010	no
stem only	0.737	0.737	0.737	0.000	no

normalisation in (3) moves it to 3.899 and 3.194 from the two class covariances.

Correcting it changed decoding by +0.005 on cohort B and −0.016 on cohort A. The estimator improved and the task did not follow, so we reject this as the explanation for the inversion while retaining the correction, which is independently right. We report it because it is the first of five cases in this work where a distribution-level quantity improved without the task following (Section 7).

6.7. The adaptation rate condition

Diagnosis The running estimate tracks whatever is currently streaming. An exponential moving average with momentum m has an effective memory of approximately $1/m$ batches, or NB/m windows at batch size N and B batches. When a protocol presents a class in contiguous blocks longer than that memory, the estimate converges to the currently streaming class, and whitening by it removes the between-class difference. Cohort A’s blocks ($\tau_{\text{blk}} = 18$) sit far inside even the fastest memory; cohort B’s ($\tau_{\text{blk}} = 309$) do not.

Prediction Slowing adaptation until the memory exceeds a class block should restore cohort B. The memory is approximately 160, 640 and 3200 windows at $m = 0.2, 0.05$ and 0.01 , so the crossing of $\tau_{\text{blk}} = 309$ falls between the first and second settings.

Every arm that uses alignment gains +0.090 to +0.098 as the memory crosses the class-block length (Table 15). Neither arm that omits alignment responds: the no-align arm moves −0.010, within its 0.0107 run-to-run spread (Table 6), and the stem-only arm reproduces 0.7375 exactly at every setting despite a 160-fold change in the adaptation rate. Because that arm is deterministic, this is an exact null rather than an unresolved one. The manipulated variable therefore acts only through the component it controls, which an unrelated confound such as optimisation dynamics or regularisation would not produce.

6.8. The condition is two-sided

We predicted, and recorded the prediction before running it, that the same change would be free on cohort A, whose blocks sit far inside even the fastest memory so that the estimate was already class-mixed. It is not (Table 16).

Table 16

The same momentum change applied to both cohorts at matched estimator version, single seed. Deltas are computed at full precision, not from the rounded cells.

Cohort	$m = 0.20$	$m = 0.01$	Δ
A ($\tau_{\text{blk}} = 18$)	0.862	0.801	−0.060
B ($\tau_{\text{blk}} = 309$)	0.631	0.723	+0.092

The optima are opposite, and error in either direction costs 0.060 to 0.092, six to nine times the noise floor. The prediction failed because it accounted only for the harm a fast estimate can do and not for the benefit it was delivering, which is tracking genuine session drift.

Proposition 1 (Admissible rate band). *A running covariance estimate with effective memory μ windows both avoids tracking the streaming class and tracks session drift only if*

$$\tau_{\text{blk}} < \mu < \tau_{\text{drift}}. \quad (4)$$

Where $\tau_{\text{blk}} \geq \tau_{\text{drift}}$ the band is empty and no setting of μ satisfies both requirements.

Cohort A satisfies (4) with a wide margin, so fast adaptation is both safe and useful there, and slowing it forfeits the drift tracking for no compensating gain. Cohort B’s 309-window blocks force μ slow enough that it can no longer track drift, so the band is empty.

Remark 1 (The residual). This accounts for the residual without a third mechanism. Even at its best setting, alignment still costs 0.060 on cohort B (0.723 against 0.783 for the no-align arm), because at the only rate that avoids class tracking there is no drift-tracking benefit left to collect. The layer then contributes only estimation noise.

Remark 2 (Why offline alignment does not meet this failure).

A whole-recording estimate, as in Euclidean Alignment (He and Wu, 2020), is the maximally slow setting. It therefore never tracks the class, and it never tracks within-recording drift either, which is exactly the capability an in-network version is introduced to add. The failure mode is created by the move that creates the benefit.

Both bounds in (4) are properties of the recording protocol. τ_{blk} is read off the trial structure, and τ_{drift} is estimated from the Riemannian distance between session covariances, which needs no labels. The condition is therefore checkable before a model is fitted.

7. Discussion

7.1. Mechanism validation is not performance validation

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity, a divergence, an alignment distance or a nuisance-decodability score, with

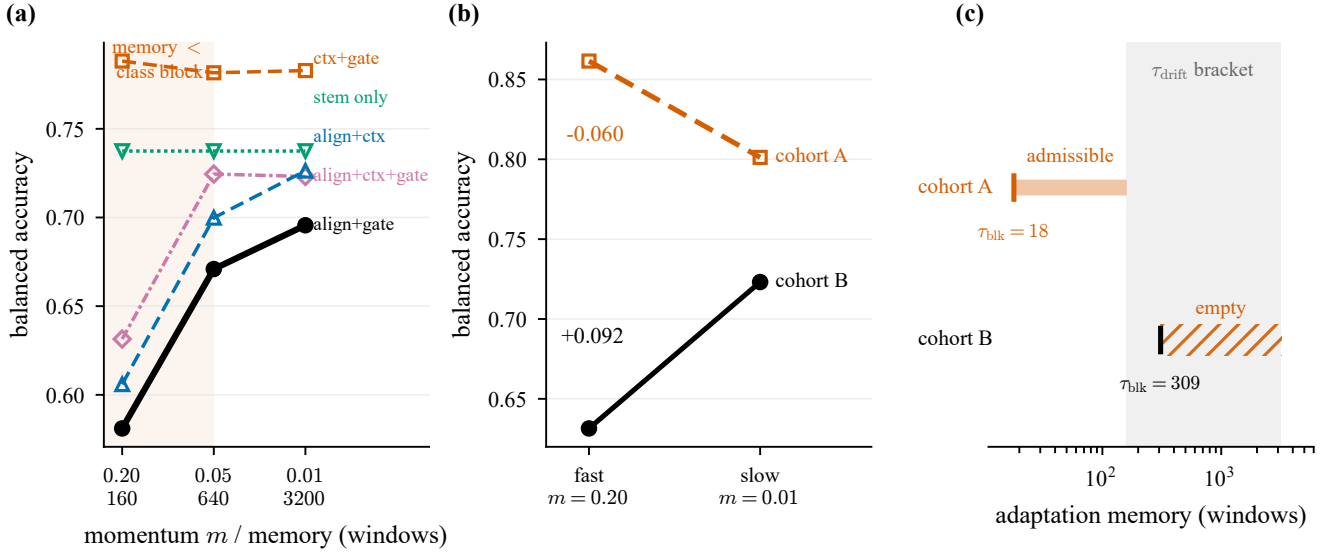


Figure 6: The adaptation rate condition. (a) Cohort B, five arms, three momenta, single seed; the shaded region is where the estimator memory is shorter than a class block. Measurement points are marked because the sweep is sparse. (b) The same change applied to both cohorts at matched estimator version, giving opposite signs. (c) The admissible band of Proposition 1, bounded below by τ_{blk} and above by τ_{drift} ; the upper bound is bracketed by cohort A being good at 160 windows and poor at 3200, not pinned.

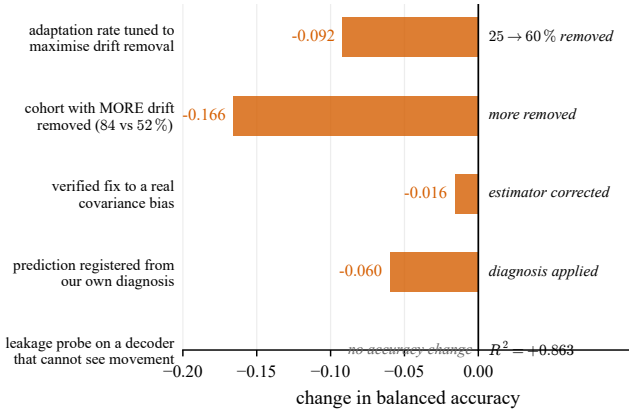


Figure 7: Five cases in which a distribution-level quantity improved (right, in blue) and the task did not follow (left, in red). The final row has no accuracy change and therefore no bar: the leakage probe rises with accuracy whether or not the decoder can access the nuisance.

task performance assumed to follow (Ganin, Ustinova, Ajakan, Germain, Larochelle, Laviolette, Marchand and Lempitsky, 2016; Long, Cao, Wang and Jordan, 2015; Sun and Saenko, 2016; Ben-David, Blitzer, Crammer, Kulesza, Pereira and Vaughan, 2010). This work provides five counterexamples within a single system, three of which were produced by acting on that assumption ourselves (Fig. 7).

1. The adaptation rate was originally selected because it maximised drift reduction (60% against 25%). That choice cost 0.092 accuracy on cohort B (Table 15).
2. The cohort in which more drift was removed (83.9% against 52.0%, Table 8) is the cohort in which accuracy fell by 0.166 (Table 13).

3. A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and -0.016 (Section 6.6).
4. A prediction registered from our own diagnosis was falsified (Table 16), because the diagnosis accounted only for the harm the estimate can do.
5. The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance, measuring $R^2 = +0.863$ for one that cannot.

Item 4 generalises the rest. Each of these quantities is one-sided by construction: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. Drift reduction measures how much non-stationarity a layer removes and has no term for what the layer destroys. A diagnosis inherited from such a metric inherits its blind spot, which is why our own threshold rule was correct about one direction and silent about the other until we tested it where it predicted no effect. The practical implication is that reporting a reduced divergence, increased invariance or lowered nuisance decodability is evidence about the representation and not evidence that the method helps, even when the reduction is large and reproducible.

7.2. Relation to temporally correlated test-time adaptation

The vision literature has identified that test-time adaptation degrades when the test stream is temporally correlated (Gong et al., 2022; Niu et al., 2023; Wang et al., 2022), and proposes reservoirs and stabilised objectives as remedies. Our result is compatible with that line and adds two things specific to the blocked-protocol case. First, the relevant quantity is a ratio of timescales rather than a qualitative

property of the stream, and the threshold location is predicted rather than fitted (Table 15). Second, the constraint is two-sided, because in a longitudinal deployment the same estimator that can track the class is also the one that tracks the drift, so the remedies that slow adaptation trade one failure for the other rather than removing both.

8. Limitations

We state the objections a reviewer should raise, because each is a real constraint on what the results support.

Single seed. The ablations are seed 0. With a 0.010 noise floor on transformer-containing arms (Table 6), single-seed rankings of closely spaced models are not interpretable, and we avoid making them: the 0.012 gap to ATCNet in Table 11 is reported as a loss rather than resolved in either direction. The effects we do interpret (+0.090 to +0.098, -0.060) are six to ten times that floor. Multi-seed replication of the headline ablation remains outstanding.

No paired significance tests on accuracy. Per-participant accuracies were not retained, only the mean and between-subject SD, so the accuracy comparisons carry dispersion but not paired tests. The drift results (Table 8) are paired and tested; the accuracy comparisons are not.

Cohort size. Seven and eight participants. This is typical for exoskeleton BCI studies (He et al., 2018) but small, and the between-subject SD of 0.06 to 0.13 (Table 10) is large relative to the differences between arms.

Best-epoch evaluation. Model selection uses the best epoch by held-out balanced accuracy, which is conventional but upward-biased on small folds. The bias applies equally to all arms and baselines within a pipeline, so comparisons within a pipeline are unaffected, but absolute values should be read as optimistic.

The upper bound is bracketed, not measured. τ_{drift} in (4) is bounded by cohort A performing well at a 160-window memory and poorly at 3200. We have not measured an interior optimum, and a cohort whose admissible band is narrow but non-empty would test Proposition 1 properly.

The rule is post-hoc. Proposition 1 was derived after observing cohort B fail. One prediction from it has since been tested out-of-sample, and that test falsified the one-sided form (Table 16), which is how the two-sided form arose. The corrected form has not yet predicted anything in advance.

Window length differs between cohorts. Cohort B uses 2 s windows against cohort A's 4 s, so the context pathway spans a shorter duration there. This is a confound for the context ablation specifically, though not for the rate condition, which is measured within each cohort.

9. Conclusion

An estimate that adapts at test time tracks whatever is currently streaming. Under a blocked experimental protocol that includes the class label, and the layer then removes the signal it was inserted to protect. The rate at which it adapts must exceed the class-block timescale and remain below

the drift timescale, and where those requirements cross no admissible rate exists. Both bounds are read from the recording protocol, which makes this a design-time check.

The immediate open question is whether the band in (4) has an interior optimum that can be located rather than bracketed. Neither cohort here can answer it: cohort A's band is wide enough that the fastest setting tested is already inside it, and cohort B's is empty. A protocol with τ_{blk} between roughly 100 and 200 windows would place the two bounds close together, and the prediction is that accuracy would peak at an intermediate momentum and fall on both sides. That experiment is pre-registrable from the protocol alone, and it is the test we would want before treating Proposition 1 as established rather than as a description of two cohorts.

A second direction follows from the residual. If a single running estimate cannot simultaneously avoid the class and track the drift, then the two requirements should perhaps be served by two estimators at different rates, with the fast one constrained to the subspace in which drift lives and the slow one supplying the class-neutral reference. Whether such a decomposition is identifiable without labels is open, and it is the version of this architecture we would build next.

Finally, the broader caution transfers beyond EEG. Five times in this system a distribution-level quantity improved while the task did not follow, and in three of those cases we had acted on the assumption that it would. Such metrics are one-sided by construction. Any method justified by a reduced divergence should report what the reduction costs, and any diagnosis inherited from one should be tested in the direction where it predicts nothing will happen.

CRedit authorship contribution statement

Author Name: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Both datasets are publicly available. All code, configurations and result files, including those of experiments that failed, are in the accompanying repository.

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